

Lecture 7

Curse of Dimensionality

Feature Engineering

Feature transformation

- Missing value imputation
- Handling Categorical Features
- Outlier Detection
- Feature scaling

Feature Construction

Feature splitting

Feature selection

dates after model

to check

whether

which features

impacts data

know

Feature extraction

PCA

and other algo like SVD

Now COD (Curse of Dimensionality)

features

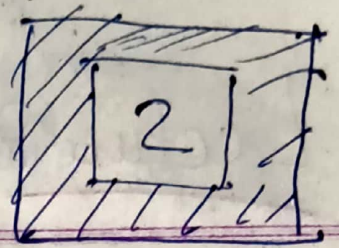
dimension

columns in data

more than optimal feature will reduce data accuracy because its irrelevant data.

low can damage
more can damage.

waste of
area



but choose optimal one

It is in high dimensional data

images & text data

but why HD data ? Sparsity

n
means high dimension data to find the data point is hard & difficult so our

means data points
more distance.

ML algo will fail.

performance
decreases
Computation time

Solution is dimensionality reduction

means reduce the no of columns

Feature
Selection

Feature
Extraction

to select subset of
feature

PCA
(Principal Component
Analysis)

1. Forward Selection

2. Backward elimination.

2) LDA
Linear Discriminant Analysis

3) t-sne